Consecutive 3-Cycle Graphs $G_3(k, n)$

Distance formulas, holography, and mismatch structure

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Main object

Binary-word state space

For integers $n \geq 3$ and $1 \leq k \leq n-1$, define

$$\mathcal{B}(k,n) = \{\text{binary strings of length } n \text{ with exactly } k \text{ zeros}\}.$$

A vertex is a configuration of k indistinguishable zeros among n positions.

Canonical source

$$A = 0^k 1^{n-k}.$$

For a target vertex B, write the zero positions as

$$1 \leq z_1 < z_2 < \cdots < z_k \leq n.$$

Consecutive 3-cycle generators

For a string $s = s_0 s_1 \cdots s_{n-1}$ and $i \in \{0, \ldots, n-3\}$, act only on the triple (s_i, s_{i+1}, s_{i+2}).

Forward cycle

$$\sigma_i^+(s): \quad (s_i, s_{i+1}, s_{i+2}) \mapsto (s_{i+2}, s_i, s_{i+1}).$$

Reverse cycle

$$\sigma_i^-(s): \quad (s_i, s_{i+1}, s_{i+2}) \mapsto (s_{i+1}, s_{i+2}, s_i).$$

$$\sigma_i^+ \quad \boxed{s_i}\ \boxed{s_{i+1}}\ \boxed{s_{i+2}} \longrightarrow \boxed{s_{i+2}}\ \boxed{s_i}\ \boxed{s_{i+1}}$$

Graph definition

Consecutive 3-cycle Schreier graph

The graph $G_3(k,n) = (V,E)$ has

$$V = \mathcal{B}(k,n).$$

Two distinct vertices $u, v \in V$ are adjacent if

$$v = \sigma_i^+(u) \quad \text{or} \quad v = \sigma_i^-(u)$$

for some consecutive triple position $i \in \{0, \dots, n-3\}$.

- Constant triples 000 and 111 are fixed and do not produce edges.
- The generators preserve the number of zeros.
- Since $\sigma_i^- = (\sigma_i^+)^{-1}$, the adjacency relation is symmetric.

Example of one move

$s = 1100$, position $i = 1$

The active triple is $(s_1, s_2, s_3) = (1, 0, 0)$. Under the forward cycle,

$$(1, 0, 0) \mapsto (0, 1, 0),$$

so

$$\sigma_1^+(1100) = 1010.$$

$$1100 \xrightarrow{\sigma_1^+} 1010$$

Displacements, area, and parity correction

Let $B \in \mathcal{B}(k,n)$ have zero positions $1 \le z_1 < \cdots < z_k \le n$. Define

$$d_i = z_i - i, \qquad I(B) = \sum_{i=1}^{k} d_i, \qquad p_i = d_i \bmod 2.$$

Alternating parity correction

$$\Delta(B) = \left| \sum_{i=1}^{k} (-1)^{i+1} p_i \right| .$$

- $I(B)$ is the area / inversion statistic relative to A.
- $\Delta(B)$ records the parity obstruction created by 3-cycle moves.

Canonical distance formula

Exact distance from $A = 0^k 1^{n-k}$

$$\boxed{d_{G_3(k,n)}(A, B) = \frac{I(B) + \Delta(B)}{2}}$$

Equivalently,

$$d(A, B) = \left\lceil \frac{I(B)}{2} \right\rceil + h(B),$$

where $h(B)$ is the holography mismatch offset.

The formula separates the metric into:

- a geometric area term $I(B)$;
- a parity correction term $\Delta(B)$.

Bounds for arbitrary pairs

For arbitrary vertices $X, Y \in \mathcal{B}(k,n)$ with ordered zero positions

$$X : x_1 < \cdots < x_k, \qquad Y : y_1 < \cdots < y_k,$$

the graph distance satisfies

$$\boxed{\left\lceil \frac{1}{2} \sum_{i=1}^{k} |x_i - y_i| \right\rceil \leq d_{G_3(k,n)}(X,Y) \leq \sum_{i=1}^{k} \left\lceil \frac{|x_i - y_i|}{2} \right\rceil}.$$

Diameter

$$\boxed{\mathrm{diam}(G_3(k,n)) = \left\lceil \frac{k(n-k)}{2} \right\rceil}$$

The maximum is attained between $0^k 1^{n-k}$ and $1^{n-k} 0^k$.

Holography rule

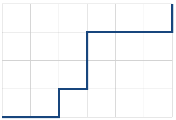
Represent a vertex B by a lattice path in a $k \times (n-k)$ rectangle:

$$0 = \text{move right}, \qquad 1 = \text{move up}.$$

Let U be the area under this path. In this setup,

$$U = I(B) = \sum_{i=1}^{k} (z_i - i).$$

ones up



zeros move right

The simplest prediction is

$$d(A,B) \approx \left\lceil \frac{U}{2} \right\rceil .$$

Mismatch condition

A vertex is a holography mismatch when

$$d(A, B) \neq \left\lceil \frac{I(B)}{2} \right\rceil .$$

Using the exact formula,

$$d(A, B) = \frac{I(B) + \Delta(B)}{2},$$

the equality with $\lceil I/2 \rceil$ holds precisely when

$$\Delta(B) = I(B) \bmod 2.$$

Mismatch criterion

$$\boxed{\Delta(B) > I(B) \bmod 2.}$$

Mismatch offset $h(B)$

Define

$$h(B) = d(A,B) - \left\lceil \frac{I(B)}{2} \right\rceil .$$

Since

$$\Delta(B) \equiv I(B) \pmod 2,$$

one obtains the universal formula

$$\boxed{h(B) = \left\lfloor \frac{\Delta(B)}{2} \right\rfloor .}$$

Consequence for small k

For $k = 3, 4, 5, 6$, every mismatch has $h(B) = 1$.

No mismatches for $k = 1$ and $k = 2$

$k = 1$

$$\Delta = |p_1| = p_1 = I \bmod 2.$$

Therefore

$$d(A, B) = \left\lceil \frac{I}{2} \right\rceil.$$

No mismatch is possible.

$k = 2$

$$\Delta = |p_1 - p_2|, \qquad I \bmod 2 = (p_1 + p_2) \bmod 2.$$

For all four parity pairs,

$$|p_1 - p_2| = (p_1 + p_2) \bmod 2.$$

No mismatch is possible.

$k = 3$: first nontrivial mismatch case

For $k = 3$,

$$S = p_1 - p_2 + p_3, \qquad \Delta = |S|.$$

The only mismatch parity pattern is

$$(p_1, p_2, p_3) = (1, 0, 1),$$

which is equivalent to all three zero positions being even.

Mismatch count

$$\boxed{m(n,3) = \binom{\lfloor n/2 \rfloor}{3}.}$$

The asymptotic mismatch fraction is

$$\frac{m(n,3)}{\binom{n}{3}} \longrightarrow \frac{1}{8}.$$

$k = 4$: all-even and all-odd families

For $k = 4$,

$$S = p_1 - p_2 + p_3 - p_4.$$

Mismatch occurs exactly for

$$(1,0,1,0) \quad \text{or} \quad (0,1,0,1).$$

These correspond to:

- all zero positions even;
- all zero positions odd.

Mismatch count

$$\boxed{m(n,4) = \binom{\lfloor n/2 \rfloor}{4} + \binom{\lceil n/2 \rceil}{4}}$$

$$\frac{m(n,4)}{\binom{n}{4}} \longrightarrow \frac{1}{8}.$$

$k = 5$ and $k = 6$: new families

$k = 5$

Mismatch families:

all-even, all-odd, 4 even $+ 1$ odd.

$$\boxed{m(n,5) = \binom{\lfloor n/2 \rfloor}{5} + \binom{\lceil n/2 \rceil}{5} + \binom{\lfloor n/2 \rfloor}{4} \lceil n/2 \rceil}$$

$k = 6$

Mismatch families:

all-even, all-odd, $5E + 1O$, $1E + 5O$.

$$\boxed{\begin{aligned} m(n,6) = &\binom{\lfloor n/2 \rfloor}{6} + \binom{\lceil n/2 \rceil}{6} \\ &+ \binom{\lfloor n/2 \rfloor}{5} \lceil n/2 \rceil + \lfloor n/2 \rfloor \binom{\lceil n/2 \rceil}{5}. \end{aligned}}$$

Mismatch summary for $k = 3, 4, 5, 6$

k	mismatch families	asymptotic fraction
3	all-even	$1/8$
4	all-even, all-odd	$1/8$
5	all-even, all-odd, $4E + 1O$	$7/32$
6	all-even, all-odd, $5E + 1O$, $1E + 5O$	$7/32$

The pattern suggests paired asymptotic behavior:

$$k = 3, 4: \ \frac{1}{8}, \qquad k = 5, 6: \ \frac{7}{32}.$$

Edge count perspective

A triple position contributes nontrivially exactly when the local block is not constant:

$$(s_i, s_{i+1}, s_{i+2}) \notin \{000, 111\}.$$

The edge count can be written as

$$|E(G_3(k,n))| = \frac{1}{2} \sum_{s \in \mathcal{B}(k,n)} \deg(s).$$

k	n	$\lvert V\rvert = \binom{n}{k}$	$\lvert E\rvert$	$2\lvert E\rvert/\lvert V\rvert$
2	4	6	12	4.00
2	5	10	27	5.40
3	6	20	72	7.20
3	7	35	147	8.40
4	8	70	360	10.29

Interpretation

- $G_3(k, n)$ is a compact model of a constrained state space with local moves.
- The canonical distance from A splits into area plus parity correction:

$$d(A, B) = \frac{I(B) + \Delta(B)}{2}.$$

- Holography works exactly when $\Delta = I \bmod 2$.
- Mismatch vertices reveal a structured parity obstruction.
- For $k = 3, 4, 5, 6$, mismatch families admit explicit counting formulas.

Consecutive 3-cycles create a distance–area duality

with a clean parity correction.

$$\boxed{d_{G_3(k,n)}(A,B) = \frac{I(B)+\Delta(B)}{2}}$$

$$\boxed{h(B) = d(A,B) - \left\lceil \frac{I(B)}{2} \right\rceil = \left\lfloor \frac{\Delta(B)}{2} \right\rfloor}$$